AP CALCULUS AB - UNIT 1

1.1 Introducing Calculus: Can Change Occur at an Instant?

Lesson Overview

- Interpret average rate of change in a context, including correct units.
- Use shrinking intervals to estimate a rate at one instant.
- Connect secant slopes, tangent slopes, and limit notation.
- Explain why an instantaneous rate is defined through nearby values rather than a zero-length interval.

Key Knowledge and Vocabulary

Average rate of change. $\dfrac{f(b)-f(a)}{b-a}$ measures output change per unit input on an interval.

Instantaneous rate. $\lim\limits_{h\to 0}\dfrac{f(c+h)-f(c)}{h}$, when this limit exists.

Geometric meaning. Secant slopes approach the slope of the tangent line.

Context and units. The units are always output units divided by input units.

Zero interval warning. $\frac{f(c)-f(c)}{c-c}$ is undefined; a limit studies nonzero intervals that shrink toward zero.

Explanation and Strategy

Step 1. Compute an average rate first. Then shorten the interval while keeping the target instant fixed.
Step 2. For a left estimate use inputs below the target; for a right estimate use inputs above the target. Agreement supports a two-sided instantaneous rate.
Step 3. A numerical estimate should include units and a contextual sentence. A symbolic answer should display the difference quotient before simplification.

Solved Examples

Worked Example: Polynomial motion

A particle has position $s(t) = t^3 - 2t$ meters. Estimate its instantaneous velocity at $t = 2$ using $[2, 2.01]$ and $[1.99, 2]$.

Solution. $s(2) = 4$. The right secant slope is $\frac{s(2.01)-s(2)}{0.01} = 10.0601$ and the left secant slope is $\frac{s(2)-s(1.99)}{0.01} = 9.9401$. Both approach 10, so the instantaneous velocity is approximately 10 m/s. In exact limit form, $\lim_{h\to 0}\frac{(2+h)^3-2(2+h)-4}{h} = 10$.

Worked Example: Interpreting units

The amount of water in a tank is $V(t)$ liters, where t is measured in minutes. Explain the meaning of $\frac{V(18)-V(12)}{6} = -3.4$.

Solution. From minute 12 to minute 18, the tank loses water at an average rate of 3.4 liters per minute. The negative sign records a decrease.

Worked Example: A radical rate

Evaluate $\lim\limits_{h\to 0}\dfrac{\sqrt{9+h}-3}{h}$ and interpret it as the instantaneous rate of $f(x) = \sqrt{x}$ at $x = 9$.

Solution. Multiply by the conjugate: $\frac{\sqrt{9+h}-3}{h} \cdot \frac{\sqrt{9+h}+3}{\sqrt{9+h}+3} = \frac{1}{\sqrt{9+h}+3}$. The limit is $\frac{1}{6}$, so the tangent slope at $x = 9$ is $\frac{1}{6}$.

Leveled Practice

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. For $s(t) = t^2 + 3t$, find the average rate of change on $[1, 4]$.

2. For $q(t) = 120 - 5t + 2t^2$, find the average rate of change on $[2, 5]$.

3. A room warms from 18.4°C at 10:00 to 21.1°C at 13:00. Find and interpret the average rate.

4. A graph has horizontal units of seconds and vertical units of meters. State the units of a secant slope.

Medium Level

Apply the idea in one or two connected steps.

5. For $s(t) = t^3 - 2t$, calculate the secant slope on $[2, 2.1]$.

6. For the same function, calculate the secant slope on $[2, 2.01]$ and compare it with the slope on $[1.99, 2]$.

7. The height data below describe a rising balloon. Estimate the instantaneous vertical velocity at $t = 7$ seconds and include units.

t (s)	6.8	6.9	6.99	7.01	7.1	7.2
$H(t)$ (m)	41.66	42.73	43.70	43.90	44.74	45.58

8. Write a limit expression for the instantaneous rate of $g(x) = x^2 - 5x$ at $x = 3$. Do not evaluate.

Hard Level

Connect representations, conditions, and justification.

9. A fish population is modeled by $P(t) = 1000 + 40t - 3t^2$. Find the instantaneous rate of change at $t = 4$ using a limit or an equivalent algebraic derivation.

10. A ball is launched upward with height $h(t) = 60t - 4.9t^2$ meters. Find its instantaneous velocity at $t = 3$ and interpret the sign.

11. The cost of producing x devices is $C(x) = 500 + 8x + 0.02x^2$ dollars. Estimate the additional cost per device near $x = 100$.

12. A student says, "The instantaneous rate at $t = 5$ is the average rate on $[5, 5]$." Identify the error and replace the statement with a correct definition.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. For $f(x) = ax^2 + bx + d$, simplify $\lim_{h \to 0} \dfrac{f(c+h) - f(c)}{h}$.

14. Let $s(t) = t^3 + kt$. The instantaneous rate at $t = 2$ is 17. Find k.

15. For $r(t) = t^4 - 3t^2 + 2t$, determine the instantaneous rate at $t = 1$ directly from the difference quotient.

Challenge Level

Synthesize, construct, generalize, or prove.

16. Construct a quadratic position function $s(t) = at^2 + bt + c$ satisfying $s(0) = 2$, $s(1) = 5$, and instantaneous velocity 7 at $t = 1$.

17. Suppose every secant slope from $(c, f(c))$ to $(c + h, f(c + h))$ equals $m + kh$ for nonzero h near 0. Prove that the instantaneous rate at c exists and find it.

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. For $f(x) = x^2 - 4x$, which expression represents the instantaneous rate at $x = 3$?

(A) $\frac{f(3)-f(3)}{3-3}$

(B) $\lim_{x\to 3} \dfrac{f(x) - f(3)}{x - 3}$

(C) $\lim_{x\to 0} \dfrac{f(x) - f(3)}{x - 3}$

(D) $\frac{f(4)-f(2)}{2}$

2. A particle has $s(t) = 5t^2 - t$. Its instantaneous velocity at $t = 2$ is

(A) 9

(B) 10

(C) 19

(D) 20

3. Which statement best explains why a limit is needed to define an instantaneous rate?

(A) Function values are always integers.

(B) A single point has no horizontal length for an average-rate quotient.

(C) Tangent lines never intersect graphs.

(D) Average rates are always zero.

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. For $P(t) = \frac{800}{1+3e^{-0.4t}}$, a calculator gives $P(4.99) \approx 568.330$ and $P(5.01) \approx 569.645$. Which is the best estimate of $P'(5)$?

(A) 0.658

(B) 6.58

(C) 65.8

(D) 131.5

5. A calculator table for F gives secant slopes from $t = 7$ of $-2.48, -2.493, -2.499$ as the interval shrinks from the right. The most defensible estimate for the instantaneous rate is

(A) -2.50

(B) -2.48

(C) 0

(D) DNE

Free-Response Practice – No Calculator

A delivery drone has altitude $A(t) = 2t^3 - 15t^2 + 36t + 8$ meters for $0 \le t \le 6$.

(a) Write and evaluate the average vertical velocity on $[1, 4]$. [2 points]

(b) Use the difference quotient to determine the instantaneous vertical velocity at $t = 4$. [3 points]

(c) Explain the difference between the answers to parts (a) and (b) in context. [2 points]

Free-Response Practice – Calculator Required

The concentration of a medication is $C(t) = \dfrac{24t}{t^2 + 9}$ mg/L for $t \geq 0$. Use a graphing calculator where requested.

(a) Estimate $C'(2)$ using the symmetric difference quotient with $h = 0.001$. [2 points]

(b) At $t = 2$, is the concentration increasing or decreasing? Justify. [2 points]

(c) Find the smallest positive time at which the instantaneous rate appears to be 0, and explain its meaning. [3 points]